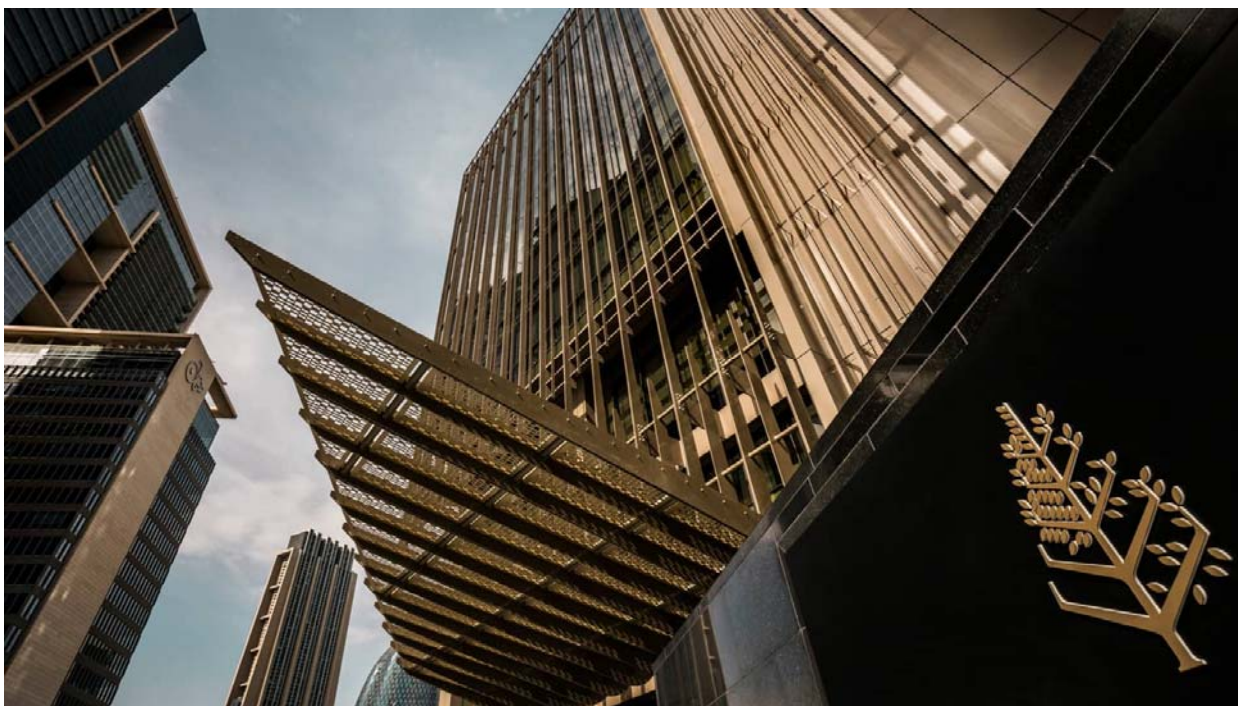




FOUR SEASONS HOTEL DUBAI INTERNATIONAL FINANCIAL CENTRE.

WASTE MANAGEMENT WALK-THROUGH AUDIT REPORT



Introduction:

Suez Middle East Recycling is the leader in providing recycling and waste management solutions in UAE and in its quest to help its Clients to comply with Dubai Municipality's waste management regulations, has embarked on a pre-site visit inspection of Four Seasons Resort(FSH) premises. The site visit was done on the 9th of February 2020. The purpose of the visit was ascertaining the



sources and types of waste generated within Four Season Resort premises and to check the availability of recycling bins to enable segregation of waste at source to comply with the Dubai municipality's mandatory source segregation regulation.

Four Season Hotel is comprised of its main kitchen, kitchen food preparation area, 2 outsourced F&B outlets, the staff pantry, Offices, common areas and one garbage room. So, these were the targeted areas during the site visit with focus on how best segregation and recycling can be mainstreamed as part of the daily operation of Four Seasons Hotel.

It is very important to note that proper segregation requires the availability of adequate recyclables receptacles both at the point of waste generation and in the Garbage room where the waste will be collected from. This is however not the case with FSH and this require appropriate action as recommended in this report.

The waste management program for FSR is broadly classified into the following properties:

1. Housekeeping waste
2. Stewarding waste
3. Outsourced F&B outlets waste (Food, Recyclable& general waste)
4. Office waste

All the properties stated above have the following scope of work:

1. General and food waste collection and disposal
2. Recycling waste collection



Considering, the dynamic nature of operations at FSH, and based on the analysis of the waste disposal provided during the audit, it is crucial to understand the gaps in the current system and streamlining waste management operations to ensure that minimal volume of waste is dumped within the landfill

To understand the gaps within the segregation system, Suez undertook the responsibility to conduct a walk-through audit to ascertain the volume of waste generated from properties within FSH, the different types of waste generated, the source of waste and the current collection streams.

Through this exercise, Suez planned to understand the gaps in recycling collection, segregation as well as internal waste movement. This process would finally help set up a process that considers the recycling resources needed, education and awareness programs needed to tackle the gap. This process will ultimately, reduce the waste ending in landfills and reduce the corresponding waste costs for haulage.

The main objective of the waste audit was as below:

- To identify the waste stream generation within FSH– with focus on various departments in FSH
- To identify the current recycling and waste resource distribution within FSH
- To identify the gaps with regards to the resources distributed and recommend appropriate resources to mainstream recycling collection
- Identify gaps in recycling awareness levels within FSH colleagues and steps to rectify the same
- To implement a measurement system that will measure progress and improve on identified areas of deficiency

METHODOLOGY:

The following methodology was followed for the waste audit at FSH:

1. Comprehensive walk through within FSH
2. Visual inspection of waste generated
3. Results
4. Recommendation and improvements based on audit findings

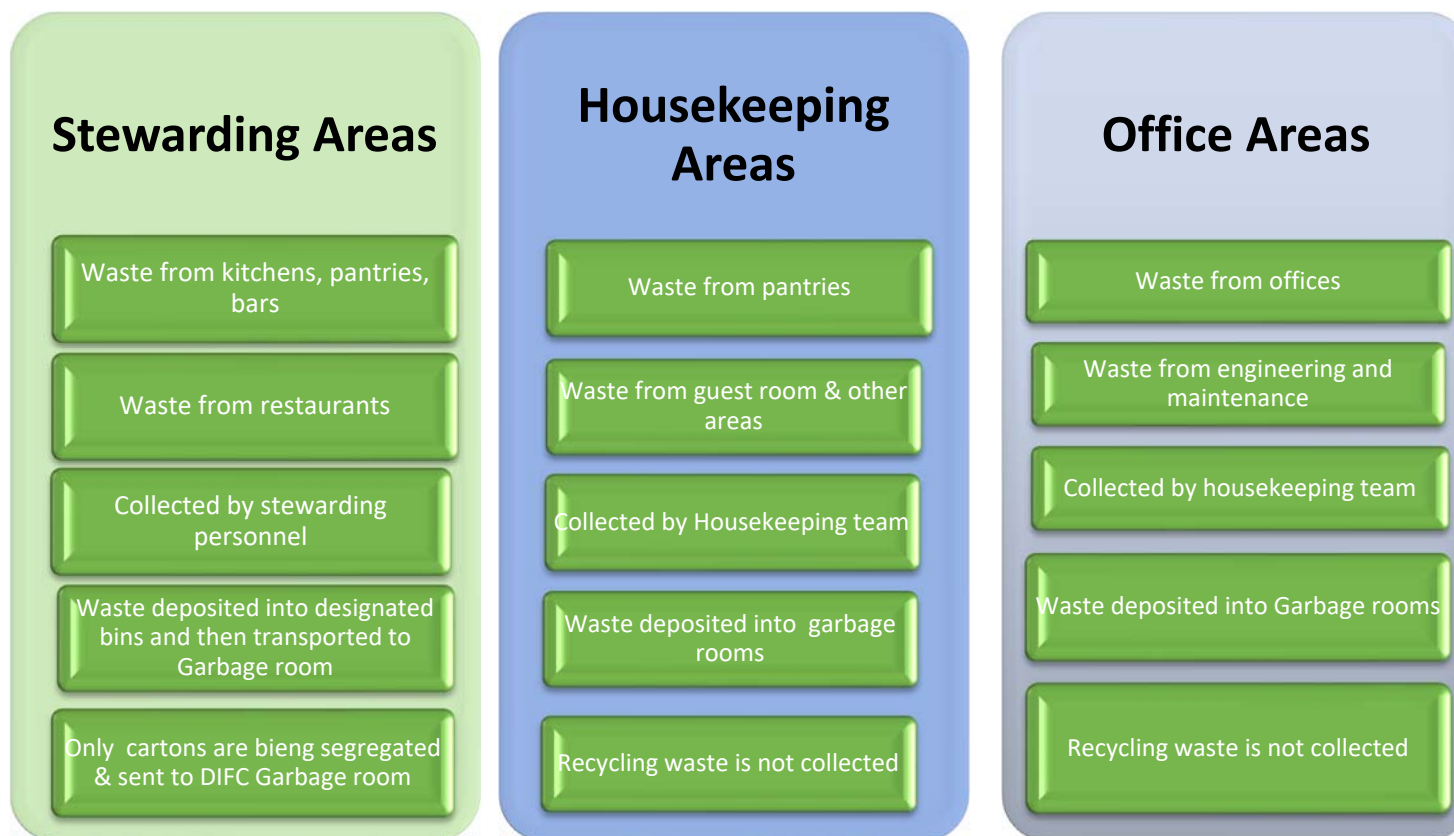


1. Comprehensive walk through FSH:

A detailed site visit was conducted to understand the internal waste collection systems, before the waste is dumped within garbage rooms.

Properties within FSH were analyzed and a walk through was done in stewarding, housekeeping and office areas to ascertain the different operations used to collect and transport the waste

The waste collection systems within FSH was analyzed with the following summary:





The waste collected from FSH had the following waste generators:

- Butler's pantry – waste generated within rooms
- Housekeeping Pantry – waste generated within rooms and from housekeeping trolleys
- Offices
- Back of the house areas
- F&B Outlets

Waste & Recycling Collection Resources:

Based on the detailed site visit, there are only general waste skips in the garbage and there is no provision for other recyclables.

There is a need for recycling bins at all housekeeping and stewarding areas for source segregation.

SUMMARY OF THE RESOURCE AUDIT:

Observations made during the Audit.

1. Currently, none of the recyclables generated within the cleaning process of the housekeeping team is recycled as the waste is not segregated.
2. All the Kitchens and food preparation stations have bins labelled for **food waste, glass, Paper& Plastic, however there is no segregation currently happening despite the availability of the bins,**
3. All other recyclables, including paper, glass, plastic bottles and cans are disposed as part of the general waste collection despite the availability of labeled bins in all



4. The waste bins within the housekeeping service areas were found with PET bottles and other recyclables mixed with general waste indicating that plastic bottles served in guest room are not being segregated.
5. There is enough space within the housekeeping service areas and currently there are four bins of different sizes on each station but still there is no segregation of recyclable paper & plastic bottles.
6. All the garbage bins in FSH have **black garbage bags** and this makes all recyclables which are segregated or partially segregated to be disposed as general waste due to lack of visibility by both the housekeeping and stewarding teams.
7. The garbage room has 1.1CBM general waste skips, and there is no provision for other recyclables like, **cartons glass, paper, plastic and cans.**
8. All cartons collected in the **1.1 skip** labelled for cartons are disposed in DIFC garbage room which leave FSH with no record of the recyclable materials generated in their premises.
9. There are **shampoos, tissues and soaps** that are left over from guest rooms which are stored for **reuse and recycling** which is a good practice than disposing them as general waste.

BOH offices:

1. The waste bins have recyclables such as paper and plastic thrown within the same.
(Please refer Annex 1 for details on the offices and recommended recycling bins)

Food and Beverage Outlets:

1. None of the food and beverage outlets practice recycling despite having designated bins for either food waste or other recyclables
2. F&B outlets and FSH' main kitchen serve glass bottled water and glass bottled soft drinks and therefore there is less plastic bottles generated.
3. The food waste in the form of fruits& Veg, rice, bread etc. is currently not segregated and therefore it disposed as part of general waste which is being sent to the landfills.
4. All the recyclables, including plastic bottles, aluminum & steel cans, are dumped within general waste.



(Please refer Annex 2 for details on the outlets and recommended recycling bins)

Identifying personnel responsible for waste collection:

Based on the on-ground analysis of the waste collection stream, the housekeeping and stewarding team undertake the responsibility of collecting the waste from bins in F&B areas and place them within the garbage room. Since recycling segregation is not practiced, all the recyclables (glass bottles, plastic bottles, metal tins, and aluminum cans) are placed as part of the general waste collection.

The internal segregation and collection process is entirely dependent on the housekeeping and stewarding crew's level of awareness on recycling and segregation. However, it was evident during the audit that there is lack of awareness for proper segregation to be done.

Type of waste generated in Four Seasons Hotel

As per the site visit and analysis, the waste generated from the FSH premises are divided into the following categories:

General and food waste:

The waste generated from kitchen, rooms, offices are a mixture of food waste and general waste, inclusive of half used tissues and paper. Considerable number of waste bags generated are mixed with general waste alongside recyclables.

Recyclable waste

Recyclables such as, glass bottles, plastic bottles, paper, and cans were primarily not collected due to the absence of the recycling bins placed within the housekeeping and stewarding areas and in the garbage room, the recyclable waste collected from the premises was dumped as general waste except for **cartons** which are sent to **DIFC garbage room carton cages**

The type of recyclables noticed during the audit, are papers, glasses, steel cans and plastic bottles. linen cloth, Soap and shampoo are also part of the reusable waste generated.

The following table gives details regarding the types of waste generated within the property and the collection and disposal process.

Observation table



SECTION NAME	TYPE OF WASTE & BIN	RECOMMENDATIONS
INDIVIDUAL OFFICES - There are 9 offices in M-level, 2 in G-floor and 1 in Basement 1 - There is no provision for paper recycling on the desks or near to the printers and all waste paper generated is placed in the general waste bin which is under the desk.		- Table top recycling bins are also required where staff can place their papers so that it will not end up in the general waste bin - Paper recycling bins should be placed at strategic centralized locations where all the papers collected from the individual offices can be disposed. So M-level with more offices can have at least 2 recycling stations. - Offices with printers should have paper recycling bins



MINA KITCHEN

- There are 6 bins
- There is no receptacle for recyclable plastic bottles & cans in the bar, and they are being disposed as general waste.
- There is a bin labelled for **bottles** but it's not specifying the type of bottles.
- There were glass bottles in the bin indicating that glasses were being segregated,
- The outlet uses black bags in all the bins

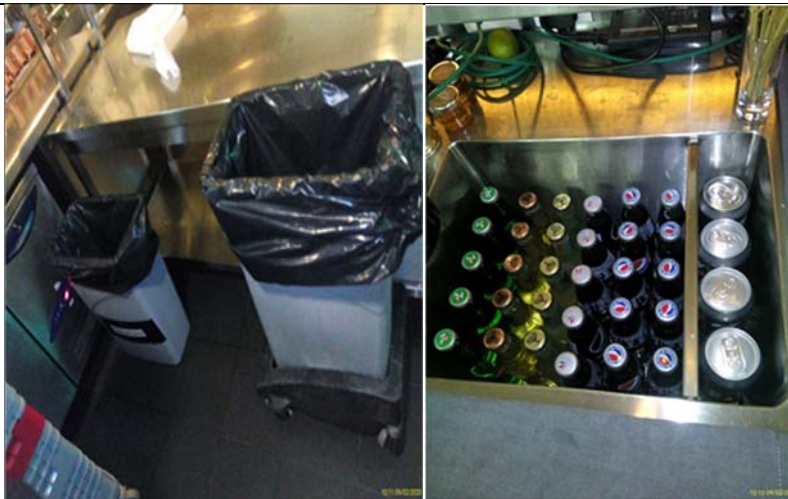


- One extra recycling bin for **plastic bottles** and cans is required in the bar so that the same can be segregated at source
- The garbage bags should be changed from **black** to **clear/transparent** bags which makes all recyclables visible.
- The label on the bin should be material specific so instead of labelling it bottles, it should be labeled **Glass**
- The label **paper and plastic** is not appropriate in the kitchen as there are rare cases of getting papers and the staff are instead, putting tissues. That bin should be labelled **Plastic bottles& Cans**



LUNA RESTUARANT

- There are 4 bins
- There is no segregation of recyclables and food waste.
- The outlet uses mostly glass bottled soft drinks and water.
- There is no bin labelled for general waste and this create confusion for staff



- All **glass bottles** and **cans** should be segregated from the general waste stream and placed in the appropriate bin as labelled.
- The bin labeled bottles should be labeled **Glass**
- One extra slim bin is required in the Bar for the **cans** as there is only a bin labeled Bottles.
- The bin which is not labelled should be labeled to cater for **general waste** to avoid confusion.
- The black bags should be replaced with clear/transparent bags to make all recyclables visible.



HOUSE KEEPING BIN STATIONS. (Butler pantry)

- There are 6 stations and each station have 4 bins, of which the big one is meant for mixed general waste and the small one is for recyclable and reusable waste like shampoos, tissues and soap.
- However, the bins are not all labeled and therefore general waste and recyclables are being mixed and disposed as general waste



- All the bins used by housekeeping should be clearly labeled to make it easy for the housekeeping team to segregate the materials before taking it to the garbage room.
- All the black bags should be replaced transparent bags
- Housekeeping team should segregate recyclables collected from guest houses particularly plastic bottles & cans
- 4-stream recycling bins with clear labels can be placed in the 6 stations to encourage segregation by housekeeping team



DISH WASHING AREA • There are 4 bins • The bin labeled for food waste is currently filled with general waste and other recyclables. • There is no bin labelled for general waste • There are not much plastic bottles generated as water and soft drinks are glass bottled.		• There are no dry papers that can be found in the dish washing area and therefore no bin should be labelled for paper & plastic. Instead the bin should be labelled Plastic bottles & Cans on to cater for steel cans from tinned foods • Black bags should be replaced with transparent bags • Food waste bin should be used for segregated food not general waste. • One extra is required and the bin should be labelled for General waste • One of the available bin should be labelled Glass for collection of glass bottles
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MAIN KITCHEN

- There are 4 bins
- There is no provision for collection of plastic bottles & cans and even some glasses where mixed with general waste.
- All the bins are labelled for paper
- There is no bin currently labeled for General waste and staff are disposing general waste in all the available bins due to lack of awareness and confusing labels.



- There is need for one extra bin for **General waste** and proper labeling of bins since the bins are of the same color
- Segregation of **steel& aluminum cans, plastic and glass** should be done to avoid sending these material to landfills
- Food waste segregation is required and one of the available bins labeled for **food waste** collection should be used for that purpose only not for general waste as shown in the picture.
- Black bags should be replaced with transparent bags and **General waste** bin should be labelled as such.
- Bin for collection of Glass bottle should be labeled **Glass** because staff can end up putting plastic bottles in the same bin.



MAIN STAFF PANTRY

- There are 3 labeled bins in which all the waste is mixed.
- There is water, coffee and soft drink dispensers, all of which uses disposable paper cups
- There is a bin labelled **paper and plastic** and this creates confusion as there are no papers generated in the pantry and they end up **disposing tissues** which fall under general waste category.
- The bin for food waste is filled with general waste.



- All the bins should be properly labelled to make it easy for staff to segregate their waste.
- **Food waste** segregation should be prioritized in the staff pantry therefore one bin should be set aside for food segregation.
- One of the available bins should be labeled for **plastic bottles and cans instead of Paper & Plastic**
- Staff should be encouraged to use **reusable mugs** instead of disposable paper cups to reduce unnecessary generation of waste.
- Only three categories should be segregated in the pantry, **Food waste, Plastic bottles & Cans, and General waste** all the bins should be labelled as such.



VEGETABLE & MPREPARATION ROOM • There are 3 bins • There is a bin labelled for Food waste collection but it's being filled with mixture of general waste and food waste.		• The bin labelled for Food waste (fruit& Veg) should be used for food waste only • There is no need to place a bin for plastic and paper in this room as only fruit & veg waste is generated. • One of the available bins should be labeled for general waste to avoid mixture with food waste.
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GENERAL WASTE	CURRENT RECYCLABLE MATERIAL COLLECTED	RECYCLABLE MATERIALS TO BE INCLUDED	COMPOSTABLE ORGANIC WASTE TO BE SEGREGATED
• Tissues • Disposable cups • General compactible waste	• Nothing is being collected for recycling except used cooking oil and cartons sent to DIFC recycling cages. • Shampoo and Soaps are stocked for reuse	• Paper • Plastic (PET bottles, LDPE, HDPE etc.) • Metal (scrap, cans etc.)	• Food waste • rice, bread, • fruit peelings • vegetables



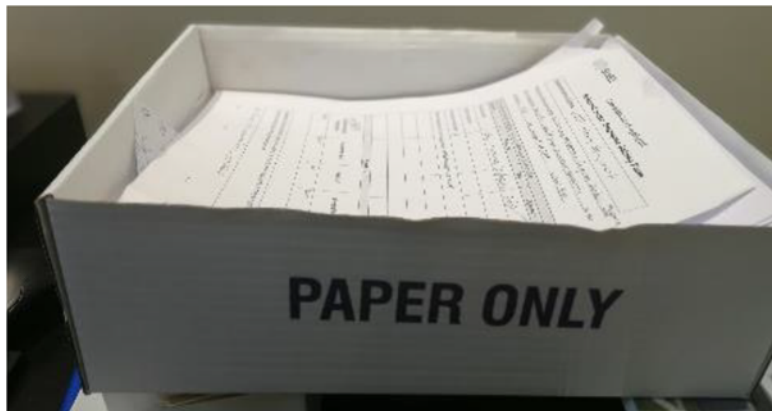
Areas for Improvement:

The following areas can be looked at for improving the current recycling processes within the FSH

OFFICES:

Recommendation for Small Offices

- A4 size table top recycling bins – with clearly instructions of 'Paper Only' can be placed for smaller offices for collection of paper.
- Staff can empty these into larger office recycling bins at the end of each day for maximum ease of use



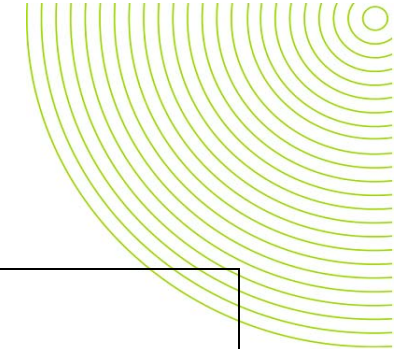
Recommendations for Big offices,

- Recycling bins targeting paper can be placed at central printing areas for collection of paper waste.
- Additionally, 3 stream recycling bins targeting, plastic, cans and glass can be placed within the pantry area or central areas within the offices, wherein the recyclables can be placed within the same.

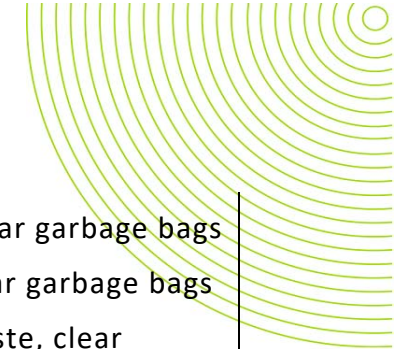
- It is recommended to place clear/transparent bin liners for the bins, so that the same can be picked up, visually observed for contamination and the recyclables can be sent to central recycling collection station within the garbage rooms.

Examples of recycling bins: (additional options can be suggested)





ANNEX 1: BOH OFFICES & RECOMMENDED RECYCLING BINS		
FLOOR OFFICES & RESIDENCES		
Office Details	Recycling Bins Needed	Remarks
Reception	1 X Table Top recycling bin for paper.	Recycling bins are not available. The paper recyclables generated are placed in the bin kept at in the reception area. However, the same is collected and dumped within the general waste collection. Water bottles generated is also part of the waste.
Housekeeping office	1 X Table Top recycling bin for paper. 1X 2 stream recycling bin	Recycling bins are not available. The paper and plastic waste generated within the offices are placed in general waste bin.
Food & beverage office	1 X Table Top recycling bin for paper.	Recycling bins are not available. The paper recyclables generated are dumped within the general waste bin col. Water bottles generated is also part of the waste.
Stewarding office	1 X Table Top recycling bin for paper.	Recycling bins are not available. The paper and plastic waste generated within the offices are placed in general waste bin.



Butler's pantry (Housekeeping-6 service area stations)	6X 4 stream recycling bins	1 bin for collection of plastic, clear garbage bags 1bin for collection of paper, clear garbage bags 1bin for collection of general waste, clear garbage bags
Engineering office	1 X Table Top recycling bin for paper. 1X 2 stream bin for paper and plastic bottles& cans	Recycling bins are not available. The paper and plastic bottles waste generated within the offices are placed in general waste bin.
HR department office	1X3 stream plastic bins, (Paper, Cans& Plastic)	1 bin for collection of plastic bottles& cans 1bin for collection of paper, clear garbage bags
Security office	1X 3 stream green, blue and red colored bin	1 bin for collection of plastic + cans, clear garbage bags 1 bin for collection of paper, clear garbage bags
FSR – Stewarding Office	1 X Table Top recycling bin for paper.	Recycling bins are not available. The paper generated within the area are placed in general waste bin.

Please Note: All the single offices that are in one floor can use one centralised 3 stream bin.



ANNEX 2: FOOD & BEVERAGE OUTLETS AND RECOMMENDED BINS

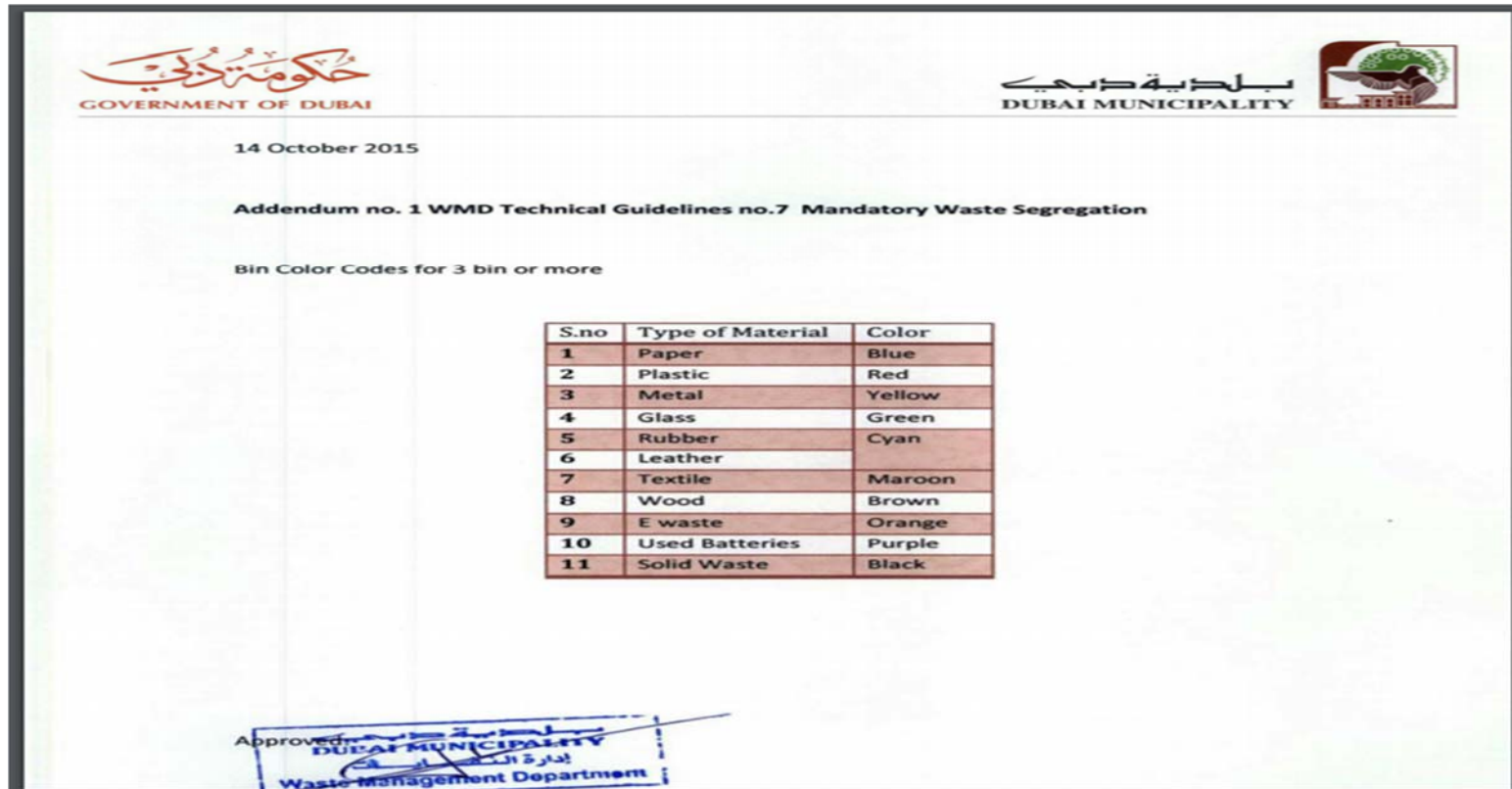
STEWARDED RESOURCE AUDIT:

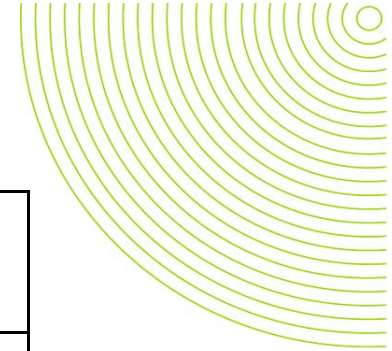
FOOD AND BEVERAGE OUTLETS		
Outlet	Recycling Bins Needed	Remarks
MINA BRASSERIE	1x30 Ltr green recycling bin in the bar	Green bin for collection of plastic and cans, clear garbage bags
LUNA RESTUARANT	2X 30ltr green and red colored bin with pedals (To be placed outside the bar)	Green bin for collection of plastic and cans, clear garbage bags Red for collection of glass, with clear garbage bags
DISH WASHING AREA	1 40ltr bin	For collection of plastic bottles, and cans, with clear garbage liners.
IN-ROOM SERVICE KITCHEN	2 stream recycling bins	For collection of plastic, cans and glass, with clear garbage liners.



FSR VEGETABLE PREPARATION ROOM	Food waste required	Available bins are adequate only labeling is requires so as not to mix food/vegetables with general waste
FSR STAFF CAFETERIA	The bin only need to be labeled for Plastic bottles & cans	There is a juicy and soft drink dispenser so there is less likely to be any plastic bottles, cans and glass 1 of the available bins can be labeled for food waste.
GUEST ROOMS	Have small general waste bins which are emptied into the main bins collected by housekeeping	Separation of waste can only be done by housekeeping before disposing in general waste bins to remove plastic bottles, cans and glass with clear garbage liners

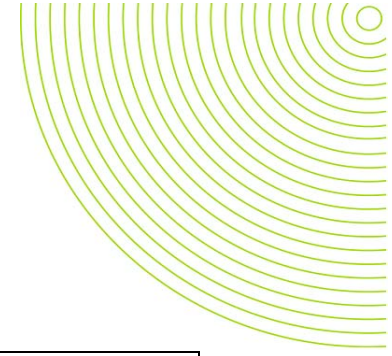
Below is the recommended color coding as per **Dubai municipality' Technical guideline No.7** for bins which are more than two particularly in the kitchens.





Steel & Aluminum cans	Recyclable Glass	General Waste (solid waste)	Plastic
			

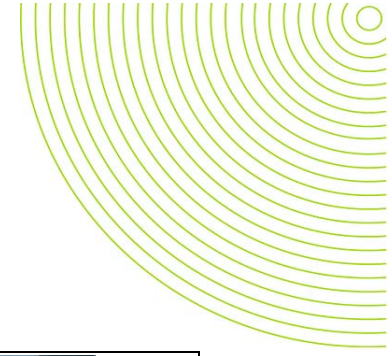
TYPE OF WASTE	EQUIPMENT USED
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RECYCLABLE CARTONS

- There currently no cage for the cartons and all cartons are disposed in one general waste skip that has been labeled for carton collection.
- All these cartons are sent to DIFC garbage room for disposal in the cages for B'eeah which means there will be no record or report for the cartons sent for recycling.
- There is no much space in the garbage room to put a bigger carton cage





USED COOKING OIL

- The barrels are placed on separate mobile platforms which seems to be smaller than the barrels and with no spill tray to contain any oil spillage on the floors
- There are three oil barrels for used cooking oil and the oil collection is done by another company.
- Two barrels with mobile spill trays are required to contain the oil spillages and make area neat and clean.





GLASS & GENERAL WASTE

- 1.1CBM Skips are available for general waste collection and waste is disposed in black bags.
- Recyclable plastic containers are being disposed as part of general waste.
- Currently there is no provision for collection of **glass bottles** and it's being disposed as general waste.
- There is need for 240litre bins where all Glasses can be disposed or one of the available 1.1cbm skips can be labeled and be used for that purpose since there seem to not much space.





FOOD WASTE COMPOSTER

- The food waste composter is currently not in use as there is no food segregation happening.
- The working condition could not be ascertained during the audit but the machine was labelled OUT OF ORDER. However, if it is working all food waste compatible with the composter can be disposed there.
- If the composter cannot handle the food waste, please refer to the below recommendation for garbage room



Recommendation for Garbage rooms

All garbage rooms should be equipped with receptacles for both general waste and recyclables

Equipment recommended include:

- 800L skip for food waste to be placed in the garbage room for segregated food waste meant for composting.
- 4X4x4 Cages for scraps, plastic bottles & cans/ in the garbage room to cater for segregated materials from housekeeping and stewarding areas and a separate labeled 240 liter bin **for glasses**



- Used cooking oil barrels and with mobile spill tray in the garbage room that can be easily accessed by all F&B crew.

Example of garbage room setup in pictures:

Cage for plastic bottles & cans



Cage for Cartons & papers.



Sample of oil barrels



Annex

3:Summary of the General waste and Recycling equipment required for the garbage room



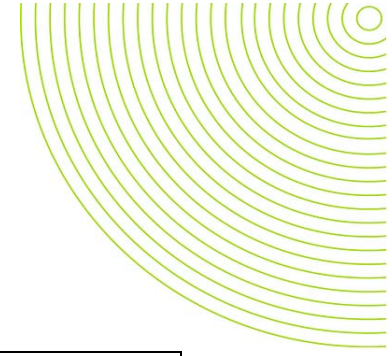
LOCATION	EQUIPMENT NAME	QUANTITY
Garbage room	Glass bins (with Wheels)	2X240litre bins
	Food waste skip 800L	1
	Recycling cage (Plastic bottles & Cans) 4x4x4	1
	Used Cooking Oil Barrel	2
	Mobile oil spill tray	1

Recommendations

- Effective Training on source segregation and sustainable development goals of Four Season Hotel is required to instill a sense of responsibility and cultivate a culture of environmental stewardships amongst the tenants so that they understand their role in waste management.
- F&B outlets are the major producers of both general and recyclable waste therefore their active participation in recycling plays a pivotal role through source segregation of their waste. All F&B outlets need to be informed that source segregation is mandatory for all waste generators as **per Dubai Municipality Technical Guideline No. 7 Mandatory waste segregation.**
- Proper labeling of the available bins is required as mentioned in the observation table to avoid confusion amongst the housekeeping and stewarding team when segregating the waste since the bins are not following the DM color coding.
- The color coding for the recycling bins as per DM has been shared as part of this report for reference when purchasing the bins.



Garbage rooms should have used cooking oil barrels that have oil spill trays which are mobile, which can be used by F&B outlets for disposing their used cooking oil to avoid keeping it in the shop floors as it compromises the hygienic standards due to floor spillages. Currently the barrels are on a flat platform with no wheels or spill tray making it difficult to transport and its vulnerable to floor spillages
NOTE: 100% food waste segregation should be done for the waste to be accepted for composting. That means no plastics, gloves, metals or any other non-biodegradable material should be mixed with food waste. If the available composter is not going to be used, 800liter Bin is require for food waste and this should be collected separately from general waste.
Suez recommends the use of clear or transparent garbage bins in place of black garbage bags for easy identification of recyclables and increase the recycling rate fast, as visible recyclables will be easily noticed and removed from the general waste stream. This will also help to identify partially segregated waste that might have been mixed with general waste. This has been tested and it proved to give excellent result in reducing the quantity of waste going to the landfills.
- Suez recommends that a circular from the Dubai Municipality requiring all commercial outlets to do source segregation and a memo from the management, should be sent to all the outsource F&B outlets so that they become aware of this legal requirement and start segregating their waste. Training on waste segregation should be compulsory for all the F&B Outlets, management representatives, stewarding and housekeeping team for recycling initiative to be successful - Suez will provide the awareness training on source segregation to ensure that all involved stakeholders are aware of their duties in sustainable waste management.
POSITIVE OBSERVATIONS



- There is already an initiative of putting labelled bins in all the food preparation areas however this need to be complimented with awareness training for the staff to ensure proper segregation.
- There is positive management commitment to start recycling and composting of waste.

Suez Middle East Recycling is committed to ensure the success of recycling initiative in Four Season Hotel DIFC and in building a sense of Environmental stewardship amongst the F&B outlets, management and housekeeping team. Maintaining a clean environment and implementing sustainable and innovative waste management practices remains SMER's top priority to keep its clients within the confines of the Dubai Municipality regulations that aims to increase landfill diversion